Exercises

Exercise 7.1 · Euclid's algorithm. Find $\gcd(1{,}071, 462)$ with Euclid's algorithm, then simplify the fraction $\frac{462}{1{,}071}$.

Solution. $1{,}071 = 2 \times 462 + 147$, $462 = 3 \times 147 + 21$, $147 = 7 \times 21$. The gcd is 21 and $\frac{462}{1{,}071} = \frac{22}{51}$.

Exercise 7.2 · Bézout pair. Find a pair of integers (u, v) such that $17u + 12v = 1$.

Solution. $17 = 12+5$, $12 = 2\times5+2$, $5 = 2\times2+1$. Working back: $1 = 5-2\times2 = 5-2(12-2\times5) = 5 \times 5 - 2 \times 12 = 5(17 - 12) - 2 \times 12 = 5 \times 17 - 7 \times 12$. The pair $(5, -7)$ works: $85 - 84 = 1$.

Exercise 7.3 · Coprime integers. Let n be a natural number. Prove that n and $n + 1$ are coprime, then that $2n + 1$ and $3n + 1$ are coprime too.

Solution. $(n + 1) \times 1 + n \times (-1) = 1$. Next $3(2n + 1) - 2(3n + 1) = 1$. Bézout's theorem concludes in both cases.

Exercise 7.4 · A homogeneous equation. Find all pairs of integers (x, y) such that $15x = 21y$.

Solution. Dividing by 3: $5x = 7y$. So $7 \mid 5x$ and $\gcd(7, 5) = 1$: Gauss gives $x = 7k$, $k \in \mathbb{Z}$. Then $5 \times 7k = 7y$, that is $y = 5k$. Conversely $15 \times 7k = 105k = 21 \times 5k$. The solutions are the pairs $(7k, 5k)$, $k \in \mathbb{Z}$.

Exercise 7.5 · Exercise books and pens. A bookshop sells exercise books at 1,100 ariary and pens at 700 ariary. A customer spends exactly 10,000 ariary. How many exercise books and pens did they buy?

Solution. We look for natural numbers x and y such that $1{,}100x + 700y = 10{,}000$, that is $11x + 7y = 100$. Modulo 7: $4x \equiv 2 \pmod 7$; since $4 \times 2 = 8 \equiv 1$, $x \equiv 4 \pmod 7$. The particular solution $(4, 8)$ works ($44 + 56 = 100$). The integer solutions are $(4 + 7k, 8 - 11k)$, $k \in \mathbb{Z}$. For $x \geq 0$ and $y \geq 0$, we need $k \geq -\frac{4}{7}$ and $k \leq \frac{8}{11}$, so $k = 0$: 4 exercise books and 8 pens.

Exercise 7.6 · Inverse modulo 26.

1. Explain why 7 has an inverse modulo 26 and check that 15 is one.
2. Solve $7x + 3 \equiv 10 \pmod{26}$.

Solution.

1. $\gcd(7, 26) = 1$; $7 \times 15 = 105 = 4 \times 26 + 1$.
2. $7x \equiv 7 \pmod{26}$; multiplying by 15: $x \equiv 105 \equiv 1 \pmod{26}$. The solutions are the integers $1 + 26k$.

Exercise 7.7 · A gcd that depends on n. Let n be a natural number and $d = \gcd(n+5, 2n+3)$.

1. Prove that d divides 7.
2. Find the integers n for which $d = 7$.

Solution.

1. d divides $2(n+5) - (2n+3) = 7$.
2. $d \in \{1, 7\}$. If $d = 7$, then $7 \mid n+5$, that is $n \equiv 2 \pmod 7$. Conversely, if $n \equiv 2 \pmod 7$, then $n+5 \equiv 0$ and $2n+3 \equiv 7 \equiv 0 \pmod 7$, so $d = 7$. Example: $n = 2$ gives $\gcd(7, 7) = 7$.

Problem

Problem · The affine cipher. The letters A to Z are numbered from 0 to 25. To encrypt a letter with number x, compute the remainder y of $7x + 3$ in the division by 26, and replace the letter by the letter with number y.

1. Encrypt the word EXPERT.
2. Prove that $y \equiv 7x + 3 \pmod{26}$ is equivalent to $x \equiv 15y + 7 \pmod{26}$.
3. Decrypt the message TDNZZ.
4. Replace 7 by a in the key. Prove that if $\gcd(a, 26) \neq 1$, two different letters may be encrypted as the same letter. How many values of $a \in \{1, \dots, 25\}$ give a cipher that can be decrypted?

Solution · Problem.

1. E= $4 \mapsto 31 \equiv 5$ (F); X= $23 \mapsto 164 \equiv 8$ (I); P= $15 \mapsto 108 \equiv 4$ (E); E $\mapsto$ F; R= $17 \mapsto 122 \equiv 18$ (S); T= $19 \mapsto 136 \equiv 6$ (G). The encrypted word is FIEFSG.
2. If $y \equiv 7x + 3$, multiplying $y - 3 \equiv 7x$ by 15: $x \equiv 15y - 45 \equiv 15y + 7 \pmod{26}$. Conversely, if $x \equiv 15y + 7$, then $7x + 3 \equiv 105y + 52 \equiv y \pmod{26}$, since $105 \equiv 1$ and $52 \equiv 0$.
3. T= $19 \mapsto 15 \times 19 + 7 = 292 \equiv 6$ (G); D= $3 \mapsto 52 \equiv 0$ (A); N= $13 \mapsto 202 \equiv 20$ (U); Z= $25 \mapsto 382 \equiv 18$ (S). The message is GAUSS.
4. If $d = \gcd(a, 26) > 1$, let $x' = x + \frac{26}{d}$. Then $ax' = ax + 26 \times \frac{a}{d} \equiv ax \pmod{26}$: the letters x and $x' \neq x$ (for x and x' in $\{0, \dots, 25\}$, for example $x = 0$) have the same encryption. If $\gcd(a, 26) = 1$, the inverse of a modulo 26 allows decryption as in 2. The $a \in \{1, \dots, 25\}$ coprime with $26 = 2 \times 13$ are the odd numbers other than 13: there are 12 of them.

Teacher note · Status of the problem. Ciphers (affine, Vigenère, Hill, RSA) are "possible problems" of the syllabus. The Hill cipher is taken up in chapter 9 (matrices), RSA in chapter 8. Have the cipher programmed in Python.